

Mario Lopez

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Work Experience

University of Southern California – Software Engineer

January 2025 – Current

- Partnered with researchers and product stakeholders to identify usability issues and improve access to critical research workflows.
- Redesigned DABI, a large research platform, using Svelte, Tailwind CSS, and Rust/Axum; improved accessibility, stability, and page load time from 5 seconds to under 1 second.
- Built secure access control with JWT authentication and HIPAA-compliant data handling for sensitive research data.
- Worked with UX/UI design teams to build a reusable component library for internal teams, improving interface consistency and reducing duplicated frontend work.
- Wrote clear technical specs for engineers and AI-assisted coding tools, reducing implementation time by about 30%.
- Added automated tests with Cypress and accessibility validation aligned to WCAG 2.1 standards.

Molecular Testing Labs – Software Engineer

August 2023 – October 2024

- Identified onboarding bottlenecks and designed HL7 integration tooling that enabled laboratory staff to independently onboard new clients, reducing onboarding time by 50%.
- Developed a secure API gateway that enabled clients to track orders and submit data, improving visibility into order workflows.
- Built a HIPAA-compliant Laboratory Information System using C#, .NET, Azure, SQL, and Kafka; designed scalable microservices and integrated NetSuite ERP to automate order management.
- Created an internal operational dashboard that displayed real-time equipment data and reduced lab downtime.
- Partnered with QA and product teams to automate regression testing using Cypress, Postman, and CI/CD pipelines.
- Mentored a junior developer on debugging, CI/CD, and deployment best practices to support team enablement.

Selected Freelance Projects – Software Engineer

February 2023 – Present

- Delivered custom websites and SEO-optimized infrastructure for small businesses, improving performance, discoverability, and user engagement.
- Refactored a React app to Next.js with SSR, cutting load times by 40% and boosting SEO rankings.
- Wrote onboarding documentation that reduced ramp-up time by 50% and reliance on senior developers.

Dolthub – Software Engineer

May 2022 – January 2023

- Built end-to-end team permissions using Go and GraphQL, implementing RBAC to improve account security and team collaboration.
- Developed HostedDolt's home and pricing pages in Next.js to improve UI consistency, page performance, and customer conversion.
- Improved SEO performance and search visibility by automating Robots.txt and Sitemap.xml generation with on-page optimizations.
- Authored engineering documentation and implementation guidelines for frontend performance optimization.

Amgen – Associate Engineer

January 2022 – May 2022

- Maintained twelve HIPAA-compliant internal applications used by more than 200 staff members.
- Fixed bugs and UX issues that reduced support tickets by 25%.
- Designed and built a secure resource transfer application in a single sprint and delivered it ahead of schedule.

HelloAdvisr – Jr. Developer / Intern

July 2020 – December 2020

- Built a full-stack pricing strategy application using React, Node.js, and Firebase, helping small businesses model and optimize revenue scenarios.

Education

University of California, San Diego

June 2021

Bachelor of Science in Computer Science

President, Triangle STEM Fraternity, 2019–2021; rebuilt dormant chapter from 0 to 6+ active members and received Outstanding Chapter President recognition.

Skills

Solutions: Requirements Discovery, Stakeholder Collaboration, Client Onboarding, API Integration, HL7 Integration Tooling, Workflow Automation, Technical Documentation, Dashboards

Technical: React, Next.js, TypeScript, JavaScript, Svelte, REST APIs, GraphQL, Go, C#, .NET, Node.js, Rust, Axum, SQL

Cloud/Tooling: Azure, Azure DevOps, Kafka, Linux, CI/CD, Git, Cypress, Postman, Jest, HIPAA, WCAG 2.1, SEO, Agile/Scrum